26900145 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEBCA302 - MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. What is a training dataset? What is its purpose in Machine Learning? [K] / [CO1]
2. What is Machine Learning? [K] / [CO1]
3. A house price is predicted using its area, number of rooms, and age. Which type of regression is suitable for this problem? Give a reason. [U] / [CO2]
4. What is meant by linearly separable data? [U] / [CO3]
5. Define a hyperplane in SVM. [U] / [CO3]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. What is a Confusion Matrix? Explain its four components with a suitable example. [K] / [CO1]

7. A dataset has many features, but only some of them are useful. Explain how feature selection can help. [A] / [CO2]

8. Explain backpropagation and optimization in neural networks. [U] / [CO3]

9. Demonstrate the working principle of BIRCH and mention its advantages. [U] / [CO4]

10. Explain agglomerative hierarchical clustering with its major steps. [U] / [CO4]

11. Describe any three ethical considerations that should be addressed while developing and deploying machine learning systems. [U] / [CO4]

12. Suppose a company wants to divide its customers into five groups based on purchasing behaviour. Explain how you would apply K-Means clustering to solve this problem. [A] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. Explain Supervised Learning, Unsupervised Learning, Semi-Supervised Learning, and Reinforcement Learning with suitable examples. [U] / [CO1]

14. Explain the steps involved in constructing a Decision Tree using a suitable example. Explain the concepts of Entropy and Information Gain and describe how they are used to select the best splitting attribute. Discuss the advantages, disadvantages, and major issues in Decision Tree learning. [An] / [CO2]

15. What is activation function? Explain any 4 activation function in detail with its mathematical formula and working. [An] / [CO3]

[15x2 = 30]